analysis. In fact, we have already encountered two examples of duality. In Chapter 10, we discussed how string theory in a universe that has a circular dimension of radius R can equally well be described as a universe with a circular dimension of radius 1/R. These are distinct geometrical situations that, through the properties of string theory, are actually physically identical. Mirror symmetry is a second example. Here two different Calabi-Yau shapes of the extra six spatial dimensions—universes that at first sight would appear to be completely distinct—yield exactly the same physical properties. They give dual descriptions of a single universe. Of crucial importance, unlike the case of English versus Chinese, there are important physical insights that follow from using these dual descriptions, such as a minimum size for circular dimensions and topology-changing processes in string theory.

In his lecture at Strings ‘95, Witten gave evidence for a new, profound kind of duality. As briefly outlined at the beginning of this chapter, he suggested that the five string theories, although apparently different in their basic construction, are all just different ways of describing the same underlying physics. Rather than having five different string theories, then, we would simply have five different windows onto this single underlying theoretical framework.

Before the developments of the mid-1990s, the possibility of such a grand version of duality was one of those wishful ideas that physicists might harbor, but about which they would rarely if ever speak, since it seems so outlandish. If two string theories differ with regard to significant details of their construction, it’s hard to imagine how they could merely be different descriptions of the same underlying physics. Nonetheless, through the subtle power of string theory, there is mounting evidence that all five string theories are dual. And furthermore, as we will discuss, Witten gave evidence that even a sixth theory gets mixed into the stew.

These developments are intimately entwined with the issues regarding the applicability of perturbative methods we encountered at the end of the preceding section. The reason is that the five string theories are manifestly different when each is weakly coupled—a term of the trade meaning that the coupling constant of a theory is less than 1. Because of their reliance on perturbative methods, physicists have